

Story-to-Motion: Synthesizing Infinite and Controllable Character Animation from Long Text

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*The sun was shining brightly, casting a warm glow over the quiet residential quarter. The air was filled with the sounds of birds chirping and leaves rustling in the gentle breeze. We decided to start our day by **dancing in front of the modest buildings**, moving to the beat of our favorite songs. As we danced, we felt the energy of the neighborhood pulsing through us. After a while, we **continued to dance around the trees**, feeling the cool breeze on our faces as we moved. It was a refreshing break from the quietness of the residential quarter, and we felt invigorated by the fresh air and natural beauty around us. As our stomachs began to growl, we **spotted a food cart selling hot dogs**.....*



Figure 1: *Story-to-Motion* is a new task that generates motions and trajectories aligned with a given story (top green area).

ABSTRACT

Generating natural human motion from a story has the potential to transform the landscape of animation, gaming, and film industries. A new and challenging task, **Story-to-Motion**, arises when characters are required to move to various locations and perform specific motions based on a long text description. This task demands a fusion of low-level control (trajectories) and high-level control (motion semantics). Previous works in character control

and text-to-motion have addressed related aspects, yet a comprehensive solution remains elusive: character control methods do not handle text description, whereas text-to-motion methods lack position constraints and often produce unstable motions. In light of these limitations, we propose a novel system that generates controllable, infinitely long motions and trajectories aligned with the input text. **1)** We leverage contemporary Large Language Models to act as a text-driven motion scheduler to extract a series of (text, position, duration) pairs from long text. **2)** We develop a text-driven motion retrieval scheme that incorporates motion matching with motion semantic and trajectory constraints. **3)** We design a progressive mask transformer that addresses common artifacts in the transition motion such as unnatural pose and foot sliding. Beyond its pioneering role as the first comprehensive solution for Story-to-Motion, our system undergoes evaluation across three distinct sub-tasks: trajectory following, temporal action composition, and motion blending, where it outperforms previous state-of-the-art (SOTA) motion synthesis methods across the board. Homepage:

<https://story2motion.github.io/>

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SA Technical Communications '23, December 12–15, 2023, Sydney, NSW, Australia
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ACM ISBN 979-8-4007-0314-0/23/12...\$15.00
<https://doi.org/10.1145/3610543.3626176>

CCS CONCEPTS

• **Computing methodologies** → **Motion capture.**

KEYWORDS

text-to-motion, motion matching, motion in-betweening

ACM Reference Format:

Zhongfei Qing, Zhongang Cai, Zhitao Yang, and Lei Yang. 2023. Story-to-Motion: Synthesizing Infinite and Controllable Character Animation from Long Text. In *SIGGRAPH Asia 2023 Technical Communications (SA Technical Communications '23)*, December 12–15, 2023, Sydney, NSW, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3610543.3626176>

1 INTRODUCTION

Imagine the potential of effortlessly translating a long textual narrative detailing a series of human activities traversing diverse locations into seamless, lifelike human motions. This transformation not only ushers in a new era of content generation but also has the power to reshape the animation, gaming, and film industries. Leveraging the capabilities of Large Language Models, this ambitious vision has now evolved from a distant goal into a tangible reality. Herein, we introduce a pioneering effort, *Story-to-Motion*, a task that takes a "story" (exemplified in Figure 1) as input and faithfully generates a sequence of motions that simultaneously conforms to low-level kinematic and high-level semantic constraints.

Previous works in character control and text-to-motion have addressed related aspects. Some efforts [Holden et al. 2020; Zhang and Tang 2022] mainly focus on matching trajectories, supporting relatively few semantic motions, while the others [Zhang et al. 2022, 2023] only focus on generating short semantic motions and ignore long motions with trajectories. Neither approach fully integrates trajectory and semantic description, making it difficult to solve the problem effectively. Besides, directly generating motions with learning-based methods is limited by motion quality, particularly for long-tail training sets. Neglected due to data imbalance, less frequent motions like handstands yield unsatisfactory results.

To generate high-quality and long motion, motion matching is widely adopted in the industry. While practical, it can not be directly applied to this problem due to its inability to utilize text input and blending artifacts. To address these challenges, we incorporate text embedding to match candidate motions. Furthermore, kinematics features and learned features are leveraged for further retrieval. Additionally, dynamic target trajectory is proposed to improve trajectory matching. Besides, heuristic-based blending algorithms yield subpar results with complex motions. To address this, we design a progressive mask transformer for motion transitions.

Our contributions are three-fold: 1) We propose a new task, *Story-to-Motion*, which considers both trajectory and semantics when generating motions. 2) We propose *Text-based Motion Matching*, a promising long text-driven, controllable system to address this task. 3) Through experimentation on standard datasets, our design outperforms the current SOTA methods in all three sub-tasks.

2 RELATED WORKS

To generate motion from a given trajectory, motion matching [Clavet 2016] retrieves motion segments based on the current pose and the target trajectory. Learned motion matching [Holden et al.

2020] employs a neural network to predict the next motion state based on the control signal. GAMMA [Zhang and Tang 2022] employs reinforcement learning to generative motion primitives via body surface markers.

Text-to-motion has witnessed notable progress. TEACH [Athanasios et al. 2022] utilizes transformer-based VAEs and is able to achieve temporal action composition. Moreover, MotionDiffuse [Zhang et al. 2022] adapt diffusion models to generate human motions from text input. ReMoDiffuse [Zhang et al. 2023] further integrates a retrieval mechanism to refine the denoising process and enhance the generalizability and diversity.

Motion blending is a variant of motion completion [Duan et al. 2021], where the resulting motion is constrained by given context frames. For deep learning methods, SOTA approaches primarily rely on transformer networks. MC-Trans [Duan et al. 2021] utilizes a pretrained language model and extra 1D convolution layers. Two-stage transformers [Qin et al. 2022] propose a two-stage approach that demonstrates improved ability by utilizing Keyframe Positional Encoding and Learned Relative Positional Encoding.

3 METHODOLOGY

3.1 Text-driven Motion Scheduler

Given a story, the Text-based Motion Scheduler module prompts ChatGPT [Brown et al. 2020] in natural language for a list of text descriptions of the character's actions $\mathcal{T} = \{text_i\}$, location names $\mathcal{L}$, and duration t of those actions (Figure 2). Assuming that the 3D scene is known, we can look up the corresponding coordinates $\mathcal{P} = \{(x_i, y_i)\}$ with $\mathcal{L}$. Then, the trajectory generation module converts the discrete $\mathcal{P}$ into a continuous curve via a path-finding algorithm. For discrete textual descriptions, each motion is additionally given a designated duration by LLM, while the remaining idle time is filled in with "walking". Thus, we transform the story into a continuous function over time, named *Scheduler* $\mathcal{S}(t)$, where each time point t corresponds to $(x_i, y_i, text_i)$, containing both low-level locations (x_i, y_i) and high-level textual descriptions $text_i$ about motion.

3.2 Text-based Motion Retrieval

Given a motion database $\mathcal{D} = \{(text_i, m_i)\}$, the *Text-based Motion Retrieval* module matches motions in an auto-regressive manner. Here, $text_i$ is the text label (e.g., "sitting") while $m_i \in \mathbb{R}^{L \times D}$ denotes a motion sequence with L frames. Each frame is a D -dimensional vector representing body joint rotation and overall translation. It is notable that the database consists of short clips. The duration of the target action is included in ChatGPT-generated instruction, which affects the number of clips. Hence, the system is scalable to arbitrary lengths of motion, as the main idea is to retrieve short clips and blend them into long motions.

At $t = 0$, a motion m_0 is randomly selected based on $text_0$ and placed at (x_0, y_0) . At the time point t , given $(x_t, y_t, text_t)$ from $\mathcal{S}(t)$ and the previous motion m_{t-1} , metrics are used to find the next best matching motion from the database. The objective is to find a motion clip that aligns with both the query text and trajectory, while also maintaining a similar body pose and motion style as the previous motion for coherence. We achieve this goal in two steps.

In the first step, to incorporate semantic information, a pre-trained sentence encoder [Liu et al. 2019] extracts text embedding

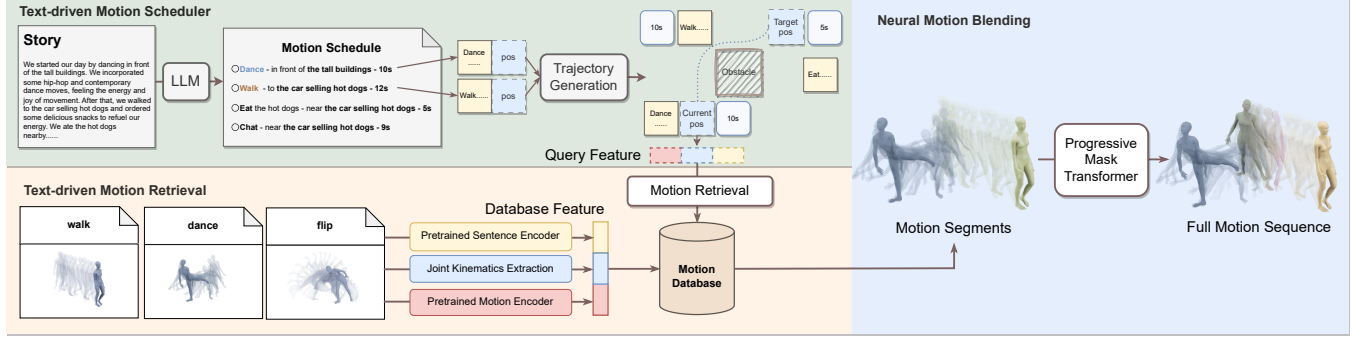


Figure 2: The proposed Text-based Motion Matching framework includes: (1) Text-Driven Motion Scheduler extracts semantic information and obtains the trajectory. (2) Text-based Motion Retrieval retrieves motions adhering to semantic and trajectory constraints. (3) Neural Motion Blending generates transition motion and concatenates motion segments for a natural sequence.

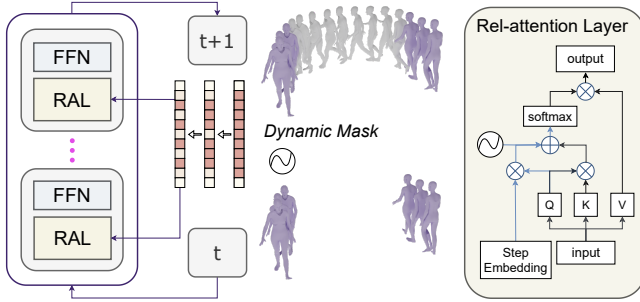


Figure 3: The Progressive Mask Transformer generates motion iteratively with decreasingly masked frames. FFN: feed-forward network. RAL: Rel-attention Layer.

f_{text} from the given text. Top- K_1 results, selected via cosine similarity, serve as candidates for subsequent matching. However, two challenges arise: (1) Imperfect text matching results in irrelevant motions. (2) Some datasets contain low-quality clips. Consequently, the presence of noisy matched results will adversely affect the quality of the generated motion. Thus outlier removal is employed to reject noisy motion clips that are far from the distribution center.

In the second step, trajectory and coherence constraints are incorporated through motion-matching. The crucial aspect is determining the similarity measure. The original Motion Matching [Holden et al. 2020] method mainly adopts trajectory and joint position similarity. The features include lower body part $f_{lower} = \{foot, \dot{foot}, hip\}$ and trajectory part $f_{traj} = \{pos, direction\}$, where pos denote the 2D future trajectory positions projected on the ground, $direction$ are the future trajectory facing directions, $foot$ are the two foot joint positions, $\dot{foot}$ are the two foot joint velocities, and hip is the hip joint velocity. However, this strategy overlooks the consistency of the upper-body, resulting in potential swaying of the upper-body. Moreover, sudden changes in motion style can negatively impact the visual quality. Therefore, we include the upper-body feature $f_{upper} = \{upper-body, \dot{upper-body}\}$ which is the upper-body joints positions and velocities. Moreover, we train an auto-encoder model $\mathcal{F}$ to extract motion features $f_{learned}$ that encompass the entire body information, together with temporal cues and motion style.

With all designed features, we compute the Euclidean similarity for each of them, including lower-body, upper-body, trajectory, and learned features. Z-Score normalizing the features is crucial due to their potential significant differences in magnitudes. To summarize, we first use text embeddings to select top- K_1 candidate motions, then the top- K_2 desired motions are selected by the weighted sum S of the above similarity, with adjustable weights for low-level and high-level scenarios. To ensure diversity, we choose clips (e.g., 10) randomly from the most suitable candidates.

Iterating the aforementioned matching process can generate arbitrarily long motions. However, its auto-regressive nature leads to cumulative trajectory errors. To address this, the target trajectory is dynamically adjusted based on the current position and target position: when a position error occurs, the subsequent step will correct this error by retrieving clips that minimize the error.

3.3 Neural Motion Blending

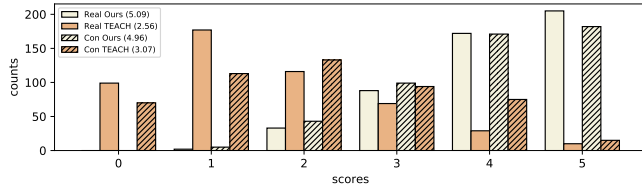
Motion matching generates a sequence of motion clips, which can be quite numerous when dealing with long text. Thus realistic transition motions is the key to high-quality result. The current SOTA method two-stage transformer [Qin et al. 2022] has two limitations: (1) It employs a full zero attention mask in the first stage, which results in unreliable information propagation when the length of the mask is large. (2) Although the two-stage design shows promising results, their parameters are not shared, which not only causes inefficiency but also prevents the second stage from fully utilizing the information learned in the first stage. We design a *Progressive Mask Transformer* (Figure 3) to tackle these issues. We propose a coarse-to-fine approach that generates motion progressively, sharing parameters among iterative steps. This approach is parameter-efficient and can leverage knowledge from the previously well-learned model. Particularly, to ensure reliable information propagation, we design a dynamic attention mask that gradually introduces more information in each iteration. Refer to Appendix A for details.

4 EXPERIMENTS

To our best knowledge, no prior work can generate infinitely long motions and trajectories aligned with the given long text. Thus we

Table 1: Motion Blending Benchmark in AMASS. The numbers in red and blue indicate the best and the second-best results.

Frames	Param	Pos / m						Rotation						Mean
		5	15	30	45	60	70	5	15	30	45	60	70	
MC-Trans [Duan et al. 2021]	13.03M	0.362	0.667	0.993	1.308	1.584	1.872	0.198	0.375	0.578	0.720	0.829	0.921	0.867
Context Trans [Qin et al. 2022]	10.39M	0.243	0.488	0.829	1.145	1.416	1.637	0.195	0.321	0.509	0.662	0.778	0.863	0.757
Detail Trans [Qin et al. 2022]	20.52M	0.142	0.398	0.762	1.082	1.358	1.576	0.123	0.265	0.465	0.624	0.744	0.828	0.697
Ours	10.39M	0.167	0.402	0.728	1.030	1.296	1.506	0.161	0.290	0.474	0.636	0.735	0.819	0.687

**Figure 4: A user study on temporal action composition (number in parentheses indicates average score).**

compare our system with SOTA techniques in trajectory following, temporal action composition, and motion blending, using the database AMASS [Mahmood et al. 2019] that unifies various datasets. Supplementary materials include visualization and details.

4.1 Trajectory Following

We compare our system with GAMMA [Zhang and Tang 2022], the current SOTA method for infinite long motion generation, and closely follow their experiment settings. As indicated in Table 2, it shows significant advantages in trajectory following (columns 4-6). GAMMA often takes a long time to reach a nearby goal with sudden speed changes or stops [Zhang and Tang 2022]. In contrast, thanks to the retrieval strategy that considers both speed and position, our method can faithfully follow the trajectory and maintain control over speed, which is crucial for generating multi-character motion and avoiding collisions. Moreover, our retrieval-based system excels in addressing penetration and floating errors (columns 1-3).

4.2 Temporal Action Composition

Temporal action composition is to generate motions aligned with a series of text descriptions, following the prescribed temporal order. We conducted a user study to compare our method with the current SOTA method TEACH [Athanasios et al. 2022]. We selected 25 subjects (6 women and 19 men), with ages ranging from 20 to 35, including animators, AI researchers, and gaming enthusiasts. As shown in Figure 4, the proposed system shows superior performance in realism and text-consistency scores, with much fewer artifacts like floating and penetration. Besides, in generating rare motions, our system excels where TEACH falls short, overcoming the significant challenges faced by deep-learning methods.

4.3 Motion Blending

In motion blending, models are evaluated on transitions spanning 5 to 70 frames. As shown in Table 1, our system shows superior performance. It achieves a 37% relative improvement over MC-Trans

Table 2: Trajectory following with speed constraints.

Trajectory	Physics Error / m			Trajectory Error / m		
	Wave	Circle	Square	Wave	Circle	Square
GAMMA	0.050	0.047	0.058	2.306	1.951	3.242
Ours	0.025	0.030	0.020	0.156	0.249	0.151

[Duan et al. 2021] (rows 1 and 4) and a 15% relative enhancement over the strong baseline [Qin et al. 2022] (rows 2 and 4). Notably, our method surpasses the previous SOTA two-stage transformers [Qin et al. 2022] (rows 3 and 4) using half of the parameters, which implies that the masked attention it used in the first stage is not optimal for long motion, as it disconnects missing frames.

5 CONCLUSION

We propose *Story-to-Motion*, generating motion and trajectory from long text. Moreover, we present *Text-based Motion Matching*, a pioneering effort that combines a large language model, motion matching, and neural blending for controllable motion generation.

Since our system is retrieval-based, its diversity is ultimately limited by the size of the motion database. Thus it is promising to combine our method with learning-based methods.

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